

Kuiyi (Cory) Gao

Chapel Hill, NC | kuiyigao@unc.edu | kuiyigao.github.io | github.com/KuiyiGao | [Google Scholar](https://scholar.google.com/citations?user=KuiyiGao)

Languages

Mandarin (native) | English (TOEFL iBT 110) | Cantonese (intermediate)

Research Focus

AI safety, agent reliability, multimodal evaluation, runtime enforcement, and human oversight.

Education

University of North Carolina at Chapel Hill

Jan 2026 to May 2027

B.A. in Computer Science

(expected)

- GPA 3.68/4.00. Coursework in algorithms, machine learning, natural language processing, technology ethics, C/Unix systems, and object-oriented design.

The Chinese University of Hong Kong

Sep 2023 to May 2026

Undergraduate study in Computer Science, Faculty of Engineering ELITE Stream

- Coursework in graduate deep learning, artificial intelligence, data structures, formal languages, computer organization, probability, statistics, and linear algebra.

Publications and Manuscripts

[1] **Kuiyi Gao*** et al. *RealGUINoise: A Dynamic Cross-Platform Benchmark for GUI Agent Robustness under Real-World Interface Noise*. Co-first author. Under review, 2026.

[2] **Kuiyi Gao**. *RuntimeTickets: Source-Grounded Security Obligations for Agent Skills*. First author. Under review at ICLR 2027.

[3] Wenxuan Wang, Xiaoyuan Liu, **Kuiyi Gao**, Jen-tse Huang, Youliang Yuan, Pinjia He, Shuai Wang, Zhaopeng Tu. *Can't See the Forest for the Trees: Benchmarking Multimodal Safety Awareness for Multimodal LLMs*. ACL 2025 Main Conference. [\[paper\]](#) [\[code\]](#)

[4] Wenxuan Wang, **Kuiyi Gao**, Youliang Yuan, Jen-tse Huang, Qiuzhi Liu, Shuai Wang, Wenxiang Jiao, Zhaopeng Tu. *Chain-of-Jailbreak Attack for Image Generation Models via Step by Step Editing*. Findings of ACL 2025. [\[paper\]](#) [\[code\]](#)

Research Experience

RealGUINoise [1] — CUHK, advised by Prof. Michael R. Lyu

Hong Kong and remote

Sep 2025 to present

- **Co-first author**. Built paired web and Android tasks with controlled overlays, dynamic content, layout shifts, and look-alike controls.
- Defined task-completion, unsafe-action, and action-sequence measures; implemented the automated judge and clean-versus-altered comparison.

RuntimeTickets [2] [\[code\]](#) — MBZUAI UGRIP, advised by Dr. Steve Liu

Abu Dhabi and Beijing

Jun 2026 to present

- **First author**. Designed task-specific contracts that narrow host permissions, gate runtime requests, and preserve decision and effect records.
- Implemented the runtime firewall in a four-student UGRIP team; lead method design, experiments, and manuscript development.

- **Second author.** Designed the multi-turn image-editing attack and prompting defense; contributed benchmark data and ran most attack and defense experiments.
- Wrote evaluation scripts and prepared paper figures, manuscript text, and rebuttal.

- **Third author.** Constructed benchmark data and evaluation for 1,500 image-text pairs across 29 safety scenarios.
- Ran vision-centric fine-tuning, analyzed missed risks and benign false alarms, and contributed to submission and rebuttal.

- Implemented a MiniLM embedding pipeline with RBF kernels, spectral entropy, centering, and covariance whitening to compare text collections without training a classifier.

Selected Projects

- **MemSlot: Long-Document Compression** (UNC, 2026). Tests whether learned saliency can compress contracts before OCR and question answering. Implemented memory-slot attention over frozen RoBERTa features, overlap-aware scoring, saliency-guided rendering, and pruning baselines; end-to-end OCR-to-QA evaluation remains incomplete. [\[code\]](#)
- **Poetry and Image Variation** (CUHK, 2025). Studies whether poetic movements differ in the visual variation produced by image models. Compared saved CLIP-based variation scores for Imagist and Romantic excerpts generated with FLUX and Stable Diffusion using SciPy and Matplotlib.
- **Movability Segmentation** (CUHK, 2025). Maps scene elements into four movability classes for action-oriented perception. Implemented DeepLabV3 with a custom ASPP head, COCO-Stuff remapping, image-mask preprocessing, and checkpoint inference. [\[code\]](#)

Honors and Awards

- Professor Joseph Lau Scholarship of Innovation; CUHK Faculty of Engineering ELITE Stream Award; CUHK College Entrance Scholarship; CUHK Exchange Scholarship for UNC, Spring 2026.
- Gaokao: top 2% in Beijing, 2023.

Technical Skills

- **Research engineering:** Python, Git, Linux, Docker, Playwright, HTTP/JSON APIs, experiment configuration, trace logging.
- **Machine learning:** PyTorch, Hugging Face Transformers, sentence-transformers, attention, fine-tuning, semantic segmentation.
- **Evaluation and analysis:** benchmark design, task and action metrics, multimodal judgment, span-QA EM/F1, kernel methods, statistical testing.